- Shapes with the same perimeter may have different areas.
- Shapes with the same area may have different perimeters.

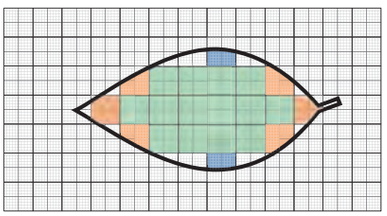
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3.5 Area of Irregular Shapes

The area of the shapes like triangle, square etc., are found by standard formulae. But we can find the approximate area of shapes like leaves as follows.



Place a leaf on a graph sheet and trace its boundary. Now observe the squares of size 1 cm x 1 cm inside of this boundary. We get complete squares (Green), partial but bigger than half squares (Orange) and half squares (Blue). The smaller than half squares which have negligible area are omitted.



Now the approximate area of the leaf

= (Number of full squares + Number of more than half squares + $\frac{1}{2}$ x Number of half squares) $sq.\ units$

= (14 + 6 + $\frac{1}{2}$ x 2) $sq.\ cm$

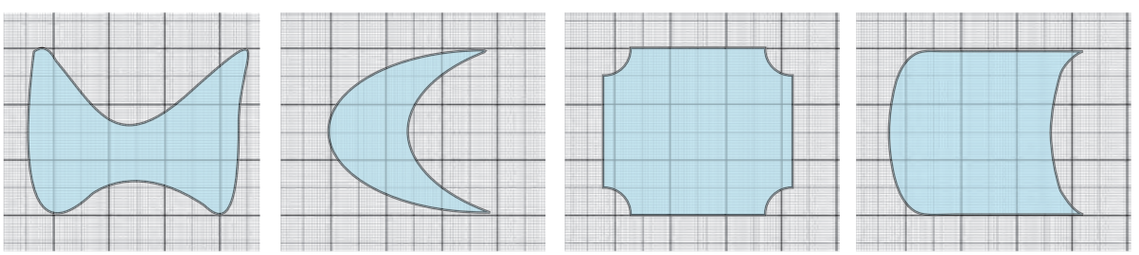
= 21 $sq.\ cm$

Note

You will learn to find the actual area of irregular shapes like leaves in your higher classes.

Try these

Find the approximate area of the following figures:



3.6 Expressing the Area in Square Units

Consider a square of side 1 cm. Therefore, its area is 1 $sq.\ cm$ (1 cm^2). Divide one of its sides into 10 equal parts. One such part is equal to 1 mm. We know that 1 cm = 10 mm. That is a square of side 1 cm is made up of 100 small squares with 1 mm square area each. Therefore, the side of this square is 10 mm and the area of this square = side x side =10 mm x 10 mm = 100 $sq.\ mm$ (100 mm^2). Therefore, the area of a square with 1 cm side is 1 cm^2 = 100 mm^2.

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